Tutorial 10

Answers

Problem 11-5

$$\begin{bmatrix} 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ -R_{12y} & R_{12x} & -R_{32y} & R_{32x} & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & R_{23y} & -R_{23x} & -R_{43y} & R_{43x} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & R_{34y} & -R_{34x} & -R_{14y} & R_{14x} & 0 \end{bmatrix} \begin{bmatrix} F_{12x} \\ F_{12y} \\ F_{32x} \\ F_{32y} \\ F_{43x} \\ F_{43y} \\ F_{14x} \\ F_{14y} \\ T_{12} \end{bmatrix}$$

$$= \begin{bmatrix} m_2 a_{g_2x} \\ m_2 a_{g_2y} \\ I_{g_2}\alpha_2 \\ (m_3 a_{g_3x} - F_{p3x}) \\ (m_3 a_{g_3y} - F_{p3y}) \\ I_{g_3}\alpha_3 - R_{p3x}\cdot F_{p3y} + R_{p3y}\cdot F_{p3x} - T_3 \\ m_4 a_{g_4x} - F_{p4x} \\ m_4 a_{g_4y} - F_{p4y} \\ I_{g_4}\alpha_4 - R_{p4x}F_{p4y} + R_{p4y}F_{p4x} - T_4 \end{bmatrix}$$

$\vec{R_{12}} = -1.414 - j1.414$,

$\vec{R_{32}} = 1.414 + j1.414$,

$\vec{R_{23}} = -4.533 - j2.11$,

$\vec{R_{43}} = 6.346 + j2.955$,

$\vec{R_{p3}} = 0$.

$\vec{R_{14}} = 2.534 - j3.095$,

$\vec{R_{34}} = 1.241 + j4.8$,

$\vec{R_{p4}} = -1.293 + j7.9$,

$\vec{a_{g_2}} = -593.9 - j537.4$,

$\vec{a_{g_3}} = -1490 - j800.355$,

$\vec{a_{g_4}} = -724.4 - j658.5$,

$\vec{F_{p3}} = 0$

$\vec{F_{p4}} = 34.6 - j20$

Problem 11-6 . using Equation 11.16c,

$$T_{12}=\frac{1}{\omega_2}\Big[m_2(a_{g_2x}\cdot V_{g_2x}+a_{g_2y}\cdot V_{g_2y})+m_3(a_{g_3x}V_{g_3x}+a_{g_3y}V_{g_3y})$$
$$+m_4(a_{g_4x}V_{g_4x}+a_{g_4y}V_{g_4y})+I_{g_2}\alpha_2\omega_2+I_{g_3}\alpha_3\omega_3+I_{g_4}\alpha_4\omega_4$$
$$-F_{p3x}V_{p3x}-F_{p3y}V_{p3y}-F_{p4x}V_{p4x}+F_{p4y}V_{p4y}-T_3\omega_3-T_4\omega_4\Big]$$

$\omega_2=20,\quad \omega_3=-5.62,\quad \omega_4=3.56$

$$\begin{cases}
\vec{v}_{g_2}=-28.284+j28.284\\
\vec{v}_{g_3}=-44.7+j31.07\\
\vec{v}_{g_4}=-11.01-j9.01\\
\vec{v}_{p3}=-44.7+j31.07\\
\vec{v}_{p4}=-38.9-j14.16\\
\vec{a}_{g_2}=-593.9-j537.14\\
\vec{a}_{g_3}=-1490-j800.355\\
\vec{a}_{g_4}=-724.4-j658.5\\
\vec{F}_{p3}=0\\
\vec{F}_{p4}=34.6-j20
\end{cases}$$

Problem 11-3

Using Equation 11.10g, with the following terms,

$$\vec{R}_{12} = -1.414 - j1.414$$

$$\vec{R}_{32} = 1.414 + j1.414$$

$$\vec{R}_{23} = -4.86 + j1.176$$

$$\vec{R}_{43} = 6.804 - j1.646$$

$$\vec{R}_{p3} = 0$$

$$\vec{a}_{g_2} = -169.7 - j113.14$$

$$\vec{a}_{g_3} = -346.8 - j132.01$$

$$\vec{a}_{g_4} = -357.17$$

$$\vec{F}_{p3} = 0$$

problem 11-4

Using equation 11.16C with the following terms

$$\vec{V_{g2}} = -14.142 + j\,14.142$$

$$\vec{V_{g3}} = -31.141 + j\,16.495$$

$$\vec{V_{g4}} = -35.140$$

$$\vec{V_{p3}} = -31.127 + j\,16.522$$

and other terms in Problem 11-3 . and use

equation 11.16C to calculate T_{12}:

$$T_{12} = \frac{1}{\omega_2}\left[\begin{array}{l} m_2\,(a_{g2x} V_{g2x} + a_{g2y} V_{g2y}) \\ + m_3\,(a_{g3x} V_{g3x} + a_{g3y} V_{g3y}) \\ + m_4\,(a_{g4x} V_{g4x}) + I_{g2}\alpha_2\omega_2 + I_{g3}\alpha_3\omega_3 \\ - F_{p3x} V_{p3x} - F_{p3y} V_{p3y} - T_3\omega_3 \end{array}\right]$$